READ-ONCE FORMULAS AND A QUESTION OF HAN ON THE FOURIER ENTROPY-INFLUENCE CONJECTURE

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ABSTRACT. For a Boolean function f and a coordinate k, let $Q_k(f) = \sum_{S \not\ni k} |\widehat{f}(S)\widehat{f}(S\cup\{k\})|/I_k(f)$, where $I_k(f)$ is the influence of k. In his proof of a new bound towards the Fourier entropy-influence conjecture, Han asked whether there is a universal constant c such that every Boolean function has a coordinate k with $Q_k(f) \le c$, and observed that this would imply the conjecture. We show that the answer is negative. For balanced alternating read-once AND-OR formulas we compute Q_k exactly by a one-step recursion over the levels of the formula; it is the same at every coordinate. Choosing the fan-ins so that every gate below the root is true with probability at most $1/4$ gives monotone read-once formulas of depth L on at most $2 \cdot 7^{L-1}$ variables with $Q_k(f) \ge L$ for every k, so $\min_k Q_k(f)$ grows at least like $\log_2 n / \log_2 7$. Depth two gives the tribes functions, for which $\min_k Q_k$ approaches 4 but stays below it. The Fourier entropy-influence conjecture is known for these formulas, so the inequality in Han's question fails on a class where the conjecture holds. By contrast, the drop of the Fourier entropy under splitting on a coordinate is less than $5 + \ln 4$ times the influence of that coordinate, for every read-once formula and every coordinate.

1. INTRODUCTION

Let $f\colon \{0,1\}^n \to \{-1,1\}$ and write $f = \sum_{S\subseteq[n]} \widehat{f}(S)\chi_S$ with $\chi_S(x) = (-1)^{\sum_{i\in S} x_i}$, under the uniform measure. The total influence is $I(f) = \sum_S |S|\widehat{f}(S)^2 = \sum_k I_k(f)$, where

$$I_k(f) = \Pr_x\big[f(x) \ne f(x \oplus e_k)\big] = \sum_{S\ni k} \widehat{f}(S)^2,$$

and the Fourier entropy is $H(f) = \sum_S \widehat{f}(S)^2 \log(1/\widehat{f}(S)^2)$. The Fourier entropy-influence conjecture of Friedgut and Kalai [1] asserts that $H(f) \le C\, I(f)$ for a universal constant C. It remains open in general; Kelman, Kindler, Lifshitz, Minzer and Safra [5] proved related statements up to a logarithmic factor, and the conjecture is known for several classes, among them read-once formulas over gates of bounded arity, by the composition theorem of O'Donnell and Tan [8], and symmetric functions and read-once decision trees, by O'Donnell, Wright and Zhou [7]. Wan, Wright and Wu [10] relate the conjecture to decision trees and communication protocols, and the inequality holds for every linear threshold function [9]. Recent work of González, MacManus and Pereyra [2] adds further classes.

Han [3] proved that $H(f) = O\big(I(f) + \sum_k I_k(f)\log(1/I_k(f))\big)$; Li and Han [6] show that this bound holds with both constants equal to 1 (with natural logarithms) for real-valued functions of unit L^2 norm. Related results are discussed by Keller, Mossel and Schlank [4]. In the course of the proof he isolates a coordinate k and the pairs of coefficients $\widehat{f}(S)$, $\widehat{f}(S\cup\{k\})$, and observes that the following inequality would lead to the conjecture by an inductive argument.

Question (Han [3, Question 3.1]). Is there a universal constant $c > 0$ such that every Boolean function f on n variables has a coordinate $k \in [n]$ with

$$\sum_{S\not\ni k} \big|\widehat{f}(S)\widehat{f}(S\cup\{k\})\big| \le c\, I_k(f)\,? \tag{1}$$

The function $x_1 \wedge \cdots \wedge x_n$ shows that $c \geq 2$ would be needed [3]. For $I_k(f) > 0$ write

$$Q_k(f) = \frac{1}{I_k(f)} \sum_{S \not\ni k} \left| \widehat{f}(S) \widehat{f}(S \cup \{k\}) \right|,$$

so that (1) asks for $\min_k Q_k(f) \leq c$, the minimum taken over the relevant coordinates.

Theorem 1.1. *For every $L \geq 1$ there is a monotone read-once formula f_L of depth L, with alternating AND and OR gates of fan-in at most 7, on $n \leq 2 \cdot 7^{L-1}$ variables, such that $Q_k(f_L) \geq L$ for every coordinate k. In particular*

$$\min_k Q_k(f_L) \;\geq\; 1 + \frac{\log_2 n - 1}{\log_2 7},$$

and the answer to Han's question is negative.

The formulas are described in Section 4, where Q_k is computed exactly for every balanced alternating read-once formula (Theorem 4.1). The computation rests on a composition identity for a single gate (Proposition 3.1), which uses only that the Fourier transform of a product of functions of disjoint sets of variables factors, and that the degree-one coefficient of a unate function is its influence. Depth one recovers Han's example, and depth two gives the tribes functions.

Proposition 1.2. *Let $T_{s,w}(x) = \bigvee_{j=1}^{s} \bigwedge_{i=1}^{w} x_{j,i}$, viewed as a $\{-1,1\}$-valued function, and $\alpha = 1 - 2^{-w}$. For every coordinate k,*

$$Q_k(T_{s,w}) = \begin{cases} 2 - 2^{2-w}, & \alpha^s \geq 1/2, \\ 4 - 2^{2-w} - 4\alpha^s, & \alpha^s < 1/2. \end{cases}$$

Hence $Q_k(T_{s,w}) < 4$ for all s, w, and $\sup_{s,w} \min_k Q_k(T_{s,w}) = 4$.

Since the Fourier entropy-influence conjecture holds for the formulas of Theorem 1.1 [8], the theorem says nothing against the conjecture itself; it shows that the route through (1) is closed, already for read-once formulas. The theorem also makes no use of the freedom to choose k: the lower bound holds at every coordinate. The entropy form of the same inductive step behaves differently on this class: the drop of the Fourier entropy under splitting on x_k is less than $(5 + \ln 4)\, I_k(f)$ for every read-once formula and every coordinate (Theorem 5.1).

Section 2 collects the facts used, Section 3 proves the composition identity, and Section 4 proves Theorem 1.1 and Proposition 1.2, Section 5 proves Theorem 5.1, and Section 6 describes exact computer checks of the closed forms against the truth tables of small formulas.

2. Preliminaries

We use $\{0,1\}$-valued functions alongside $\{-1,1\}$-valued ones. For $g\colon \{0,1\}^V \to \{0,1\}$ we write $\widehat{g}(S) = \mathbb{E}[g\chi_S]$ as well, and associate with g the function $1 - 2g\colon \{0,1\}^V \to \{-1,1\}$, whose coefficients are

$$\widehat{1-2g}(\emptyset) = 1 - 2\mathbb{E}g, \qquad \widehat{1-2g}(S) = -2\widehat{g}(S) \quad (S \neq \emptyset). \tag{2}$$

Lemma 2.1. *$Q_k(-f) = Q_k(f)$, and Q_k is unchanged when an input variable x_i is replaced by $1 - x_i$.*

Proof. Both operations multiply each $\widehat{f}(S)$ by a sign, by -1 in the first case and by $(-1)^{[i \in S]}$ in the second. Neither the influences nor the absolute values in Q_k change. □

A function is *unate in k* if it is nondecreasing in x_k or nonincreasing in x_k.

Lemma 2.2. *If $f\colon \{0,1\}^n \to \{-1,1\}$ is unate in k, then $|\widehat{f}(\{k\})| = I_k(f)$.*

Proof. Write $x^{(b)}$ for x with x_k set to b. Then $\widehat{f}(\{k\}) = \frac{1}{2}\mathbb{E}_x\big[f(x^{(0)}) - f(x^{(1)})\big]$. The difference takes values in $\{0,2\}$ or in $\{0,-2\}$, according to the direction of monotonicity, and it is nonzero exactly when k is pivotal at x. So $|\widehat{f}(\{k\})| = \Pr[f(x^{(0)}) \neq f(x^{(1)})] = I_k(f)$. □

Lemma 2.3. *Let V_1, V_2 be disjoint sets of variables, u a function of the variables in V_1 and v a function of those in V_2. For $A \subseteq V_1$ and $B \subseteq V_2$, $\widehat{uv}(A \cup B) = \widehat{u}(A)\,\widehat{v}(B)$. If v is $\{0,1\}$-valued then $\sum_{B \subseteq V_2} \widehat{v}(B)^2 = \mathbb{E}v$.*

Proof. $\chi_{A\cup B} = \chi_A \chi_B$ and the two factors of $uv\,\chi_{A\cup B}$ are independent. The second statement is Parseval's identity and $v^2 = v$. □

3. One gate

Proposition 3.1. *Let V_1, V_2 be disjoint sets of variables and $k \in V_1$. Let $g\colon \{0,1\}^{V_1} \to \{0,1\}$ be unate in k with $I_k(1-2g) > 0$, and let $G\colon \{0,1\}^{V_2} \to \{0,1\}$ have $\beta = \mathbb{E}G > 0$. Put $\mu = \mathbb{E}g$,*

$$h = (1-g)\,G, \qquad p = \mathbb{E}h = (1-\mu)\beta, \qquad f_1 = 1-2g, \qquad F = 1-2h.$$

Then $I_k(F) = \beta\, I_k(f_1)$ and

$$Q_k(F) = Q_k(f_1) + 2(1-\mu) - |1-2\mu| - 2p + |1-2p|. \tag{3}$$

Proof. Write $u = 1-g$, and let A range over subsets of $V_1 \setminus \{k\}$ and B over subsets of V_2. Put $I = I_k(f_1)$. By (2) and Lemma 2.2, $|\widehat{u}(\{k\})| = |\widehat{g}(\{k\})| = I/2$, and $\widehat{u}(\emptyset) = 1-\mu$, $\widehat{u}(A) = -\widehat{g}(A)$ for $A \neq \emptyset$.

By (2) and Lemma 2.3, $\widehat{F}(A \cup B \cup \{k\}) = -2\,\widehat{u}(A \cup \{k\})\widehat{G}(B)$ for all A, B, and $\widehat{F}(A \cup B) = -2\,\widehat{u}(A)\widehat{G}(B)$ unless $A = B = \emptyset$, while $\widehat{F}(\emptyset) = 1-2p$. Hence

$$I_k(F) = \sum_{A,B} 4\,\widehat{u}(A \cup \{k\})^2 \widehat{G}(B)^2 = 4\beta \sum_A \widehat{g}(A \cup \{k\})^2 = \beta I,$$

using $\sum_B \widehat{G}(B)^2 = \beta$ and $I = \sum_A \widehat{f_1}(A \cup \{k\})^2 = 4\sum_A \widehat{g}(A \cup \{k\})^2$.

For the numerator $N(F) = \sum_{S \not\ni k} |\widehat{F}(S)\widehat{F}(S \cup \{k\})|$, the pair $A = B = \emptyset$ contributes $|1-2p| \cdot 2|\widehat{u}(\{k\})|\beta = |1-2p|\,I\beta$, and every other pair contributes $4|\widehat{u}(A)\widehat{u}(A \cup \{k\})|\widehat{G}(B)^2$. Summing the latter over all pairs and removing the pair $A = B = \emptyset$,

$$N(F) = 4\beta \sum_A |\widehat{u}(A)\widehat{u}(A \cup \{k\})| - 4(1-\mu)\tfrac{I}{2}\beta^2 + |1-2p|\,I\beta.$$

On the other hand, by (2),

$$N(f_1) = |1-2\mu|\,I + 4\sum_{A \neq \emptyset} |\widehat{g}(A)\widehat{g}(A \cup \{k\})|,$$

$$4\sum_A |\widehat{u}(A)\widehat{u}(A \cup \{k\})| = 2(1-\mu)I + 4\sum_{A \neq \emptyset} |\widehat{g}(A)\widehat{g}(A \cup \{k\})|,$$

so $4\sum_A |\widehat{u}(A)\widehat{u}(A \cup \{k\})| = N(f_1) + 2(1-\mu)I - |1-2\mu|I$. Substituting, and using $(1-\mu)\beta = p$,

$$N(F) = \beta\big(N(f_1) + 2(1-\mu)I - |1-2\mu|I\big) - 2p\,\beta I + |1-2p|\,\beta I.$$

Dividing by $I_k(F) = \beta I$ gives (3). □

4. Alternating read-once formulas

Let $s_1, \dots, s_L \geq 1$ be integers. Define $\{0,1\}$-valued functions $T_0, \dots, T_L$ by letting T_0 be a single variable and T_l the NOR of s_l copies of T_{l-1} on disjoint sets of variables,

$$T_l = \prod_{j=1}^{s_l} (1 - T_{l-1}^{(j)}).$$

Thus T_L is a read-once formula on $n = s_1 \cdots s_L$ variables, and every variable is read at depth L. Each T_l is unate in every variable: T_0 is, and a NOR of unate functions of disjoint variables is unate, with the direction reversed. Let $p_l = \mathbb{E}T_l$, so that

$$p_0 = \tfrac{1}{2}, \qquad p_l = (1 - p_{l-1})^{s_l} \quad (1 \leq l \leq L).$$

Replacing the inputs by their negations at every other level turns T_L, or its negation, into a monotone formula with alternating AND and OR gates, and by Lemma 2.1 this does not change Q_k.

Theorem 4.1. *For every coordinate k of $f = 1 - 2T_L$,*

$$Q_k(f) = 2L - 1 - 4\sum_{l=1}^{L-1} p_l - 2p_L + |1 - 2p_L|.$$

If $p_L \leq 1/2$ this equals $2L - 4\sum_{l=1}^{L} p_l$.

Proof. Fix k. For each $l \geq 1$ the variable k lies in exactly one of the s_l copies of T_{l-1} that make up T_l; call it g, and let G be the product of $1 - T_{l-1}^{(j)}$ over the other copies, so that $\beta = \mathbb{E}G = (1 - p_{l-1})^{s_l - 1} > 0$ and $T_l = (1 - g)G$. Write q_l for $Q_k(1 - 2T_l)$, where T_l is the gate at level l above k. Proposition 3.1 applies, with $\mu = p_{l-1}$ and $p = p_l$, and gives

$$q_l = q_{l-1} + 2(1 - p_{l-1}) - |1 - 2p_{l-1}| - 2p_l + |1 - 2p_l|,$$

together with $I_k(1 - 2T_l) > 0$ for the next step. At level 0 the function is $1 - 2x_k$, whose only coefficient besides $\widehat{\ }(\{k\})$ is $\widehat{\ }(\emptyset) = 0$, so $q_0 = 0$. Summing over $l = 1, \dots, L$, the sum telescopes:

$$q_L = 2L - 2\sum_{l=0}^{L-1} p_l - \sum_{l=0}^{L-1} |1 - 2p_l| - 2\sum_{l=1}^{L} p_l + \sum_{l=1}^{L} |1 - 2p_l|,$$

and with $p_0 = 1/2$ this is the stated formula. □

Proof of Proposition 1.2. With $L = 2$, $s_1 = w$ and $s_2 = s$, T_2 is the NOR of s copies of $\bigwedge_i (1 - x_{j,i})$. After the substitution $x \mapsto 1 - x$ it is the negation of $T_{s,w}$, so by Lemma 2.1 $Q_k(T_{s,w}) = Q_k(1 - 2T_2)$. Here $p_1 = 2^{-w}$ and $p_2 = \alpha^s$, and Theorem 4.1 gives $Q_k = 3 - 2^{2-w} - 2\alpha^s + |1 - 2\alpha^s|$, which is the stated formula. Both branches are below 4. Taking $w \to \infty$ and $s = 4^w$, so that $\alpha^s \leq e^{-2^w} \to 0$, gives values tending to 4. □

With $L = 1$ and $s_1 = n$ the same theorem gives $Q_k = 2 - 2^{2-n}$ for $x_1 \wedge \cdots \wedge x_n$, Han's example.

Proof of Theorem 1.1. Choose s_l, for $l = 1, \dots, L$ in turn, as the least integer $s \geq 1$ with $(1 - p_{l-1})^s \leq 1/4$. Then $s_1 = 2$ and $p_1 = 1/4$. We claim that $3/16 < p_l \leq 1/4$ and $s_l \leq 7$ for $2 \leq l \leq L$. Suppose $3/16 < p_{l-1} \leq 1/4$ (true for $l = 2$). By the choice of s_l, $p_l \leq 1/4$, and $(1 - p_{l-1})^{s_l - 1} > 1/4$, so $p_l > (1 - p_{l-1})/4 \geq 3/16$. Moreover $1 - p_{l-1} < 13/16$ gives $(13/16)^{s_l - 1} > 1/4$, that is $s_l - 1 < \log 4 / \log(16/13) < 6.7$, so $s_l \leq 7$.

Since $p_L \leq 1/4$, Theorem 4.1 gives, for every k,

$$Q_k(f_L) = 2L - 4\sum_{l=1}^{L} p_l \;\geq\; 2L - L = L,$$

where $f_L = 1 - 2T_L$, made monotone with alternating AND and OR gates as described above. Finally $\log_2 n = \log_2 2 + \sum_{l\ge 2} \log_2 s_l \le 1 + (L-1)\log_2 7$. □

Remark 4.2. The fan-ins chosen in the proof are $2, 5, 6, 7, 6, 6, \dots$, and Q_k grows by about 1.13 per level: at depth 80 the formula has $\log_2 n \approx 209$ and $Q_k \approx 90.2$, so $Q_k \approx 0.43 \log_2 n$ there. Conversely, the formula of Theorem 4.1 always gives $Q_k \le 2L$, and when every fan-in is at least 2 the formula has at least 2^L variables, so $Q_k \le 2\log_2 n$ on all of these formulas.

5. The entropy split

In this section log is the natural logarithm. For $f\colon \{0,1\}^n \to \{-1,1\}$, a coordinate k and $b \in \{0,1\}$, let $f_{k\to b}$ be the restriction of f to $x_k = b$, a function of the other $n-1$ variables, and put

$$D_k(f) = H(f) - \tfrac{1}{2}\big(H(f_{k\to 0}) + H(f_{k\to 1})\big).$$

Since $I_j(f) = \frac{1}{2}\big(I_j(f_{k\to 0}) + I_j(f_{k\to 1})\big)$ for $j \ne k$, an inequality $D_k(f) \le K\, I_k(f)$ at one coordinate of every nonconstant function of a class closed under restrictions gives $H(f) \le K\, I(f)$ on that class, by induction on the number of variables. This is the inductive scheme behind (1) [3]; González, MacManus and Pereyra [2] isolate the inequality $D_k(f) \le K\, I_k(f)$ itself and prove it for several classes, while for Boolean functions in general, and for monotone ones, it is open. On read-once formulas it holds at every coordinate, although (1) fails there.

Theorem 5.1. *Let f be a read-once formula over AND, OR and NOT gates of any fan-in, viewed as a $\{-1,1\}$-valued function, and let k be a variable read by f. Then*

$$D_k(f) < (5 + \log 4)\, I_k(f).$$

Consequently $H(f) \le (5+\log 4)\, I(f)$ for every such formula.

The consequence follows from the induction above, since a restriction of a read-once formula is a read-once formula or a constant. The proof of the inequality has three parts: an expression for D_k through pairs of coefficients, a one-gate identity parallel to Proposition 3.1, and an estimate of the function that weighs the pairs.

Let $\eta(x) = x\log(1/x)$ for $x > 0$, $\eta(0) = 0$, and for $l, m \ge 0$

$$\Phi(l,m) = \eta(l) + \eta(m) - \tfrac{1}{2}\Big[\eta\big((\sqrt{l}+\sqrt{m})^2\big) + \eta\big((\sqrt{l}-\sqrt{m})^2\big)\Big].$$

Lemma 5.2. $D_k(f) = \sum_{S \not\ni k} \Phi\big(\widehat{f}(S)^2, \widehat{f}(S\cup\{k\})^2\big)$. *The function Φ is symmetric, $\Phi(l,0) = 0$, and $\Phi(\lambda l, \lambda m) = \lambda\, \Phi(l,m)$ for $\lambda > 0$. In particular $D_k(-f) = D_k(f)$, and D_k is unchanged when an input variable is replaced by its negation.*

Proof. For $S \not\ni k$ the character $\chi_{S\cup\{k\}}$ equals χ_S at $x_k = 0$ and $-\chi_S$ at $x_k = 1$, so $\widehat{f_{k\to 0}}(S) = \widehat{f}(S) + \widehat{f}(S\cup\{k\})$ and $\widehat{f_{k\to 1}}(S) = \widehat{f}(S) - \widehat{f}(S\cup\{k\})$. With $a = \widehat{f}(S)$ and $b = \widehat{f}(S\cup\{k\})$ the squares of these two numbers are $(|a|+|b|)^2$ and $(|a|-|b|)^2$ in some order, and summing over S gives the expression for D_k. Symmetry and $\Phi(l,0) = 0$ are immediate. Homogeneity follows from $\eta(\lambda x) = \lambda\eta(x) + \lambda x \log(1/\lambda)$ and $l + m = \frac{1}{2}\big[(\sqrt{l}+\sqrt{m})^2 + (\sqrt{l}-\sqrt{m})^2\big]$. The last statement holds because D_k depends only on the squares of the coefficients, as in Lemma 2.1. □

Proposition 5.3. *In the setting of Proposition 3.1, let $I = I_k(f_1)$ and $c = 1 - 2\mu$. Then*

$$D_k(F) = \beta\Big[D_k(f_1) - \Phi(c^2, I^2) + \Phi\big(4(1-\mu)^2, I^2\big)\Big] - \beta^2\, \Phi\big(4(1-\mu)^2, I^2\big) + \Phi\big((1-2p)^2, \beta^2 I^2\big).$$

Proof. Keep the notation of the proof of Proposition 3.1. For $(A,B) \ne (\emptyset,\emptyset)$ the coefficients found there and the homogeneity of Φ give

$$\Phi\big(\widehat{F}(A\cup B)^2, \widehat{F}(A\cup B\cup\{k\})^2\big) = 4\,\widehat{G}(B)^2\, \Phi\big(\widehat{u}(A)^2, \widehat{u}(A\cup\{k\})^2\big),$$

while the pair $A = B = \emptyset$ gives $\Phi\big((1-2p)^2, 4\widehat{u}(\{k\})^2\beta^2\big) = \Phi\big((1-2p)^2, \beta^2 I^2\big)$, since $\widehat{G}(\emptyset) = \beta$ and $\widehat{u}(\{k\})^2 = I^2/4$. Summing the displayed expression over all pairs, removing the pair $A = B = \emptyset$ and using $\sum_B \widehat{G}(B)^2 = \beta$,

$$D_k(F) = 4\beta \sum_A \Phi\big(\widehat{u}(A)^2, \widehat{u}(A \cup \{k\})^2\big) - 4\beta^2\, \Phi\big((1-\mu)^2, I^2/4\big) + \Phi\big((1-2p)^2, \beta^2 I^2\big).$$

By (2), $\widehat{f_1}(A)^2 = 4\widehat{u}(A)^2$ for $A \neq \emptyset$ and $\widehat{f_1}(A \cup \{k\})^2 = 4\widehat{u}(A \cup \{k\})^2$ for all A, while $\widehat{f_1}(\emptyset) = c$ and $\widehat{f_1}(\{k\})^2 = I^2$. Hence, by homogeneity,

$$4 \sum_A \Phi\big(\widehat{u}(A)^2, \widehat{u}(A \cup \{k\})^2\big) = D_k(f_1) - \Phi(c^2, I^2) + \Phi\big(4(1-\mu)^2, I^2\big),$$

and $4\Phi\big((1-\mu)^2, I^2/4\big) = \Phi\big(4(1-\mu)^2, I^2\big)$. Substituting gives the claim. □

Lemma 5.4. *Let f and k be as in Theorem 5.1. There are $L \geq 0$, $\beta_1, \ldots, \beta_L \in (0,1]$ and $\nu_1, \ldots, \nu_L \in (0,1)$ such that the numbers defined by $I_0 = 1$, $c_0 = 0$, $D_0 = 0$ and, for $1 \leq l \leq L$,*

$$I_l = \beta_l I_{l-1}, \qquad c_l = 1 - 2\nu_l\beta_l,$$
$$D_l = \beta_l\Big[D_{l-1} - \Phi(c_{l-1}^2, I_{l-1}^2) + \Phi(4\nu_l^2, I_{l-1}^2)\Big] - \beta_l^2\, \Phi(4\nu_l^2, I_{l-1}^2) + \Phi(c_l^2, I_l^2),$$

satisfy $I_L = I_k(f)$ and $D_L = D_k(f)$.

Proof. Let $T_0 = x_k$, and let $T_1, \ldots, T_L$ be the $\{0,1\}$-valued subformulas computed at the AND and OR gates of fan-in at least two on the path from x_k to the output, in order. Each subformula of a read-once formula is unate in every variable it reads, and since there are no constants among the inputs every subformula has expectation in $(0,1)$. At the l-th of these gates the child containing x_k computes T_{l-1} or $1 - T_{l-1}$, according to the parity of the NOT gates between, and the other children compute functions of disjoint sets of variables. If the gate is an AND gate, write its value as $(1-\tilde{g})G_l$, where $\tilde{g}$ is 1 minus the value of that child and G_l is the product of the values of the other children. If it is an OR gate, write 1 minus its value as $(1-\tilde{g})G_l$, where now $\tilde{g}$ is the value of that child and G_l the product of 1 minus the values of the other children. In both cases $\tilde{g} \in \{T_{l-1}, 1 - T_{l-1}\}$, so $1 - 2\tilde{g} = \pm(1 - 2T_{l-1})$, and $(1-\tilde{g})G_l$ is T_l or $1 - T_l$, so $1 - 2(1-\tilde{g})G_l = \pm(1 - 2T_l)$. Put $\beta_l = \mathbb{E}G_l \in (0,1]$ and $\nu_l = 1 - \mathbb{E}\tilde{g} \in (0,1)$.

By induction on l, $I_k(1 - 2T_l) = I_l > 0$, $D_k(1 - 2T_l) = D_l$ and $\widehat{\ }(\emptyset)^2 = c_l^2$ for $1 - 2T_l$. At $l = 0$ the function $1 - 2x_k$ has the coefficients 0 at $\emptyset$ and 1 at $\{k\}$ and no others, so $I_0 = 1$, $c_0 = 0$ and $D_0 = \Phi(0,1) = 0$. For the inductive step apply Propositions 3.1 and 5.3 with $g = \tilde{g}$ and $G = G_l$; then $1 - \mu = \nu_l$, $c^2 = (1 - 2\mathbb{E}\tilde{g})^2 = c_{l-1}^2$ and $p = \nu_l\beta_l$, and by Lemma 5.2 the signs do not matter. Finally $f = \pm(1 - 2T_L)$. □

Lemma 5.5. *For $0 < t \leq 1$,*

$$t\big(4\log 2 + \log(1/t)\big) \leq \Phi(1,t) \leq t\big(3 + \log(1/t)\big).$$

Consequently $\Phi(l,m) \leq m\big(3 + \log_+(l/m)\big)$ for $l \geq 0$ and $m > 0$, where $\log_+ = \max(\log, 0)$.

Proof. Let $b = \sqrt{t}$. Then $\Phi(1,t) = t\log(1/t) + \rho(b)$ with $\rho(b) = (1+b)^2\log(1+b) + (1-b)^2\log(1-b)$, and $\rho(1) = 4\log 2$. With $\log(1+b) = \sum_{j\geq 1} a_j b^j$, $a_j = (-1)^{j+1}/j$, and $a_0 = a_{-1} = 0$, the coefficient of b^j in $(1+b)^2\log(1+b)$ is $e_j = a_j + 2a_{j-1} + a_{j-2}$, and the odd powers cancel in ρ, so that $\rho(b) = 2\sum_{i\geq 1} e_{2i} b^{2i}$ for $0 \leq b < 1$. Here $e_2 = 3/2$ and, for $i \geq 2$,

$$e_{2i} = \frac{2}{2i-1} - \frac{1}{2i-2} - \frac{1}{2i} < 0$$

by the strict convexity of $1/x$. Hence $\rho(b) \leq 3b^2$, which is the upper bound, and $\rho(b)/b^2$ is nonincreasing on $(0,1)$, so by continuity $\rho(b)/b^2 \geq \rho(1) = 4\log 2$, which is the lower bound.

For the second statement we may assume $l > 0$. If $l \geq m$, homogeneity and the upper bound with $t = m/l$ give $\Phi(l, m) = l\,\Phi(1, m/l) \leq m\big(3 + \log(l/m)\big)$. If $l < m$, then $\Phi(l, m) = m\,\Phi(1, l/m) \leq 3m$, because $t(3 + \log(1/t))$ is increasing on $(0, 1]$ and equals 3 at $t = 1$. □

Proof of Theorem 5.1. Take the numbers of Lemma 5.4 and put $r_l = D_l/I_l$ and $\Psi_l = \Phi(c_l^2, I_l^2)/I_l$. Dividing the recursion for D_l by $I_l = \beta_l I_{l-1}$ gives

$$r_l - \Psi_l = r_{l-1} - \Psi_{l-1} + (1 - \beta_l)\,\frac{\Phi(4\nu_l^2, I_{l-1}^2)}{I_{l-1}}.$$

Since $r_0 = \Psi_0 = 0$ and $1 - \beta_l = (I_{l-1} - I_l)/I_{l-1}$,

$$\frac{D_k(f)}{I_k(f)} = \sum_{l=1}^{L} \big(I_{l-1} - I_l\big)\,\frac{\Phi(4\nu_l^2, I_{l-1}^2)}{I_{l-1}^2} + \frac{\Phi(c_L^2, I_L^2)}{I_L}. \tag{4}$$

Let $\gamma(x) = 3 + \log 4 + 2\log(1/x)$, a decreasing function on $(0, 1]$. Since $4\nu_l^2 \leq 4$ and $I_{l-1} \leq 1$, Lemma 5.5 gives $\Phi(4\nu_l^2, I_{l-1}^2)/I_{l-1}^2 \leq \gamma(I_{l-1})$, and so the l-th term of the sum is at most $\int_{I_l}^{I_{l-1}} \gamma(x)\,dx$. With $a = I_L$ the sum is therefore at most

$$\int_a^1 \gamma(x)\,dx = (5 + \log 4)(1 - a) + 2a \log a.$$

Since $c_L^2 \leq 1$, Lemma 5.5 also gives $\Phi(c_L^2, I_L^2)/I_L \leq a\big(3 + 2\log(1/a)\big)$. Adding,

$$\frac{D_k(f)}{I_k(f)} \leq 5 + \log 4 - (2 + \log 4)\,I_k(f) < 5 + \log 4.$$ □

Remark 5.6. On the formulas of Theorem 1.1 the ratio Q_k is at least the depth at every coordinate, while $D_k/I_k < 6.39$. The constant $5 + \log 4 = 6.386\ldots$ is not sharp. If $\nu_l = 1/2$ for every l, the sum in (4) is a Riemann sum for

$$\int_0^1 \frac{\Phi(1, x^2)}{x^2}\,dx = \frac{\pi^2}{2} = 4.9348\ldots.$$

If every gate on the path from x_k to the output is an AND gate followed by a NOT gate, with $\beta_l = 1 - \delta$ (the other children can realize this value up to an arbitrarily small error, since the maps $q \mapsto q/2$ and $q \mapsto (1+q)/2$ produce every dyadic rational in $(0, 1)$ as the expectation of a read-once formula), then ν_l tends to $1/(2 - \delta)$, and the values of D_k/I_k given by Lemma 5.4 increase to $\pi^2/2$ as $\delta \to 0$; for instance 4.9250 at $\delta = 0.1$ and 4.93480 at $\delta = 0.001$. A search over finite sequences (β_l) found no value above $\pi^2/2$. We do not know whether $\pi^2/2$ is the supremum of $D_k(f)/I_k(f)$ over read-once formulas.

6. Exact check

The script `q31_check.py` computes Q_k from the truth table. With $F = 2^n \widehat{f}$, the integer Walsh-Hadamard transform of f,

$$Q_k(f) = \frac{\sum_{S \not\ni k} |F(S)F(S \cup \{k\})|}{\sum_{S \ni k} F(S)^2},$$

a ratio of integers, which the script compares with the closed forms as exact rationals. It checks the AND function for $n \leq 8$, every tribes function $T_{s,w}$ with $2 \leq sw \leq 20$ against Proposition 1.2, and 22 formulas T_L with $n \leq 20$ and depth up to 4 against Theorem 4.1; at every coordinate for $n \leq 14$, and at the first and last coordinates above that. All 60 comparisons agree. The script `deep_trees.py` evaluates the closed form along the family of the proof of Theorem 1.1 up to depth 80.

For Section 5, `q31_tree_entropy.py` and `q31_readonce_check.py` compare D_k and I_k given by Lemma 5.4 with their values from truth tables, on 15 alternating formulas and on 60 random read-once formulas of depth at most 5 with children of unequal sizes (463 coordinates); the largest difference is $2.6 \cdot 10^{-15}$. The script `q31_readonce_bound.py` checks (4) at 400 digits on 240 random sequences (β_l) and the bound of Theorem 5.1 on 323 truth-table coordinates (largest ratio 4.42), `q31_lemma_a_analytic.py` checks Lemma 5.5, and `q31_readonce_pi2.py` and `q31_readonce_beam.py` compute the values quoted in the last remark of Section 5.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

Anthropic Claude AI systems were used to develop the mathematical derivations, write and run the numerical checks, perform internal consistency and literature reviews, and write and prepare this manuscript and its publication files. The author selected the problem and directed the work; the author did not carry out the mathematics or the writing. Responsibility for submission or publication rests with the author. The AI-assisted internal reviews, compilation checks and rendered-page inspections are not external peer review or formal verification.

Data and code availability

The checks of Section 6 are produced by the scripts in the `code` directory of the repository

`github.com/yspennstate/han-question-read-once-formulas`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [11].

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